

Foundation (Jalapeno)

Q1. What is the expansion of $(x + 3)^2$?

- (A) $x^2 + 6x + 9$
- (B) $x^2 + 3x - 9$
- (C) $x^2 - 6x + 9$
- (D) $x^2 - 3x - 9$
- (E) $x^2 + 9$

Q2. Which of the following is a perfect square trinomial?

- (A) $x^2 + 4x + 4$
- (B) $x^2 - 4x - 4$
- (C) $x^2 + 2x + 1$
- (D) $x^2 - 2x + 1$
- (E) $x^2 + 1$

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
- (B) $(x - 5)^2$
- (C) $(x + 2)^2$
- (D) $(x - 2)^2$
- (E) $(x + 1)^2$

Q4. The stock price of a company is given by the expression $x^2 - 6x + 8$. Is this a perfect square trinomial?

- (A) Yes
- (B) No
- (C) Maybe
- (D) Only on weekdays
- (E) Only on weekends

Q5. What is the expansion of $(2x - 3)^2$?

- (A) $4x^2 - 12x + 9$
- (B) $4x^2 + 12x - 9$
- (C) $4x^2 - 12x - 9$
- (D) $4x^2 + 12x + 9$
- (E) $4x^2 - 9$

Q6. An investor has x dollars in a savings account. The interest earned is given by the expression $x^2 + 4x + 4$. What is the factored form of this expression?

- (A) $(x + 2)^2$
- (B) $(x - 2)^2$
- (C) $(x + 1)^2$
- (D) $(x - 1)^2$
- (E) $(x + 4)^2$

Q7. A tax consultant uses the expression $x^2 - 8x + 16$ to calculate the tax owed. Is this a perfect square trinomial?

- (A) Yes
- (B) No
- (C) Maybe
- (D) Only for high-income clients
- (E) Only for low-income clients

Q8. The expansion of $(x - 2)^2$ is

- (A) $x^2 - 4x + 4$
- (B) $x^2 + 4x + 4$
- (C) $x^2 - 4x - 4$
- (D) $x^2 + 4x - 4$
- (E) $x^2 - 4$

Open Ended Questions (FRQ)

Q9. Factor the expression $x^2 + 12x + 36$. Show all work and explain your reasoning.

Q10. The price of a stock is given by the expression $x^2 - 10x + 25$. Is this a perfect square trinomial? Explain why or why not and provide the factored form if applicable.

Chef's Tips

- Read each question carefully before answering
- Show all work and explain reasoning for free response questions
- Use only a pencil to complete the test